

Incident Report

Case Title: Ransomware Incident Investigation

Date of Report: 01/31/2026

Reported By: Security Operations Center (SOC)

Analyst: Nizar Aderbaz

Severity: Critical

1. Executive Summary

On August 24, 2016, we found that the workstation **we8105desk** (IP: **192.168.250.100**) had been infected with **ransomware**. A review of **Splunk** and **Sysmon** logs showed that **the incident began when a user accidentally downloaded a malicious file called **mhtr.jpg** from **solidaritedeproximityte.org**.**

Upon opening the file, it silently executed a hidden script (**121214.tmp**) through **Windows scripting tools**. This resulted in the encryption of the user's local files and then propagated to **an attached server (192.168.250.20), where additional files were encrypted.**

Our analysis revealed that the attack was conducted by **Cerber Ransomware**, which resulted in **the encryption of 663 documents.**

2. Incident Timeline

Timestamp	Event
T0	Initial web request to solidaritedeproximityte.org via HTTP GET.
T1	Malicious payload mhtr.jpg (disguised as an image) downloaded to the workstation.
T2	Sysmon identifies process creation for wscript.exe executing a temporary file 121214.tmp .
T3	Parent Process ID (PPID 3968) initiates encryption routines.
T4	Local encryption begins on we8105desk , targeting 406 user text files in Bob Smith's profile.
T5	Outbound SMB traffic (Port 445) detected from workstation to file server we3269srv (192.168.250.20) .
T6	Remote encryption completed: 257 unique PDF files on the file server are encrypted.

3. Indicators of Compromise (IOCs)

IP Address	Purpose / Context	Status
192.168.250.100	Victim workstation (we8105desk)	Compromised
192.168.250.20	Target file server (we3269srv)	Impacted (lateral)
92.222.104.182	Hosting solidaritedeproximityte.org	Malicious

Indicator	Type	Context
solidaritedeproximityte.org	Domain	Malicious hosting & C2
mhtr.jpg	File	Malicious payload (Cerber downloader)
121214.tmp	File	Obfuscated execution script
MIRANDA_PRI	USB media	Identified volume associated with activity

4. Impact Analysis

- **Workstation Impact:** 406 user .txt files **encrypted** within the **Bob Smith's** Windows profile. (4 initial system-related files were excluded as artifacts).
 - **Server Impact:** 257 unique .pdf files **encrypted** on the central file server via unauthorized SMB modification.
 - **Malware Variant:** **Cerber Ransomware** (identified by **encryption behavior** and **file renaming patterns**).
-

5. Attack Techniques (MITRE ATT&CK Mapping)

- **Initial Access (TA0001):** drive by compromise (T1204.002, malicious link and file).
 - **Execution (TA0002):** command and scripting interpreter (T1059.005, visual basic).
 - **Lateral Movement (TA0008):** remote services (T1021.002, SMB and windows admin shares).
 - **Impact (TA0040):** data encrypted for impact (T1486).
-

6. Root Cause Analysis

- **Weak inbound filtering controls:** The network allowed the download of an executable script disguised with a **.jpg extension** from **an untrusted domain**.
- **Unnecessary Permissions:** The workstation had write access to **a wide range of PDF directories** on the file server, **allowing the ransomware to spread laterally**.
- **Scripting Engine Abuse:** wscript.exe and cscript.exe were permitted **to run unsigned scripts from temporary directories**.

7. Recommendations

- **Application Whitelisting:** Control **wscript.exe** and **cscript.exe** to only permitted administrative scripts.
 - **Network Segmentation:** Enforce **stricter SMB signing** and **control access to workstations** on only required server shares (**Least Privilege**).
 - **Web Filtering:** Filter newly identified domains and use “**Content Disarm and Reconstruction**” (CDR) on downloaded media content.
 - **Endpoint Protection:** Use **EDR (Endpoint Detection and Response)** to stop processes with quick file rename activity.
-

8. Conclusion

The analysis confirms a successful **Cerber ransomware attack** triggered by a **drive-by download**. By tracing the **PPID 3968** and **SMB traffic**, we mapped **the spread** from a **single workstation** to the **corporate file server**. **Total data loss includes 663 documents**. Immediate remediation means **limiting which scripts can run** and **locking down shared folders on the server** so **only authorized users can access them**.

